



TEST & MEASUREMENT

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functioning," explains Nirmalkumar.

"We simulate the running conditions in a lab and then capture the readings," explains Verma.

Testing these modules, however, raises the question of quantity. Lexus LIT IS, for example, is one-of-a-kind vehicle with about 42,000 LEDs. These LEDs in 2017 model of Lexus LIT IS are programmed to change colour in response to human gestures and music. People's fascination aside, testing the electronics is already a humongous task with such devices.

Automated testing is the way to go. Testing of automotive systems has been growing in size and complexity over the past few years. A modern luxury car may very well have up to 100 engine control units. At such high numbers, regular testing just does not cut it, and calls for appropriate test platforms to address the need for automation, automotive specific I/O and flexibility.

Automation has become increasingly valuable to automotive applications due to the complexity involved in testing whole infotainment systems. Testing an infotainment system without automation may take weeks and many man-hours. An automated infotainment test system, however, does it much faster. Test time can now be reduced to a day or two at most. Test procedures can be run in parallel for multiple software stacks for highly-aggressive development.

Safety and security are important

Current systems employ RF based protocols for detection and authentication of owners to unlock doors and start the engine. These protocols allow remote eavesdropping and have done so in the past. Due

to the large code size and complexity, these extend the range of remote attacks to many tens of metres, or even kilometres.

The megamos crypto transponder, commonly found in keys and key fobs for wireless entry systems, is supposed to stop an engine from starting without the transponder being near the vehicle. However, fake frequency modulation transmitters broadcasting radio data system traffic message channel (RDS-TMC) information, adversely influencing the navigation system in cars, have been found to attack.

Tyre-pressure-monitoring systems also employ RF protocols to send pressure sensor information from within the tyre to an engine control unit in the vehicle. These protocols have been sniffed and spoofed to fool the engine control unit into reporting a false tyre-pressure warning to the driver. An attack on the system at the wrong time could have safety consequences in case the driver is distracted or alarmed.

The need for smart(er) cars

Smartcars still lack reliable means of communication. With vehicles, even a minuscule chance of error could lead to major disasters. Human involvement in driving provides for swift action being taken in real time. Autonomous vehicles leave a lot of open areas for question. A smart vehicle runs on algorithms, which is to create a parallel to the human mind. However, unlike the human mind, any number of algorithm or security-breach problems can lead to disaster. Tesla or Google cars' accidents are prime examples. Proper testing can help root out problem areas, so that the next Uber you take is much safer. **EFY**